

# Developer Knowledge Hub

Demo architecture notes - synthetic sample document

## Overview

Developer Knowledge Hub is a personal technical knowledge workspace. A developer uploads text-based PDF documents and asks questions against their own document library. Answers are grounded in retrieved passages and include source citations.

The web interface never accesses the database, Redis, object storage, or the AI provider directly. It communicates only with the Express API. The API owns authentication, authorization, document metadata, retrieval, and AI orchestration.

## Document processing

After an upload, the API stores the PDF in private object storage, records a PENDING document, and places an identifier-only job on BullMQ. A worker reads the stored file, extracts text, splits it into bounded chunks, creates embeddings, and writes vectors to PostgreSQL with pgvector.

The visible lifecycle is PENDING, PROCESSING, CHUNKS\_READY, EMBEDDING, then READY. FAILED is a terminal state. Manual retry is available only from FAILED and restarts processing while preserving the document identifier. Deletion hides the document immediately and completes cleanup asynchronously.

## Grounded answers and source trust

For each question, the API retrieves relevant chunks only from the signed-in owner's non-deleted READY documents. Conversation history can clarify follow-up wording, but it is not factual evidence. The answer receives newly retrieved, source-labelled passages and returns citation markers such as [S1] and [S2].

The application validates citation references and persists the response together with citation snapshots. Source cards identify the document and page. If evidence is insufficient, the assistant must acknowledge that rather than inventing a source.

## Security and operating boundaries

Sign-in uses revocable PostgreSQL sessions in HttpOnly cookies. Backend queries enforce owner scope, so one developer cannot retrieve another developer's documents or conversations. Uploaded PDFs and extracted text are treated as untrusted input.

This is a portfolio demonstration document with synthetic content only. It contains no credentials, private user records, or operational secrets.